a

FY-2D, FY-2E, INSAT-3A and INSAT-3D viewed the aircraft most directly; MTSAT-2 most obliquely

View zenith angle at the aircraft (deg)

Satellite (sub-satellite longitude)
- FY-2D (86.5° E)
- FY-2E (104.1° E)
- INSAT-3A (93.4° E)
- INSAT-3D (82.0° E)
- Kalpana-1 (73.9° E)
- Elektro-L N1 (76.0° E)
- COMS-1 (128.2° E)
- Meteosat-7 (57.6° E)
- MTSAT-2 (145.0° E)

b

Effective infrared pixel size (km)

c

Effective visible pixel size (km)

Line and shading key
- Posterior routes, weighted median
- Posterior routes, 5-95 % range
- Military radar track (to 18:01:49)
- Scan reached the aircraft (median)
- Night (Sun below -6 deg)
- Civil twilight

d

Time the scan reached the aircraft

e

Elevation at the aircraft (deg)
- Sun elevation
- Moon elevation

Time (UTC), 7-8 March 2014

Plain note: This chart shows how each weather satellite in a fixed position above the equator saw the aircraft during the night of 7-8 March 2014, how large one image pixel was at the aircraft, and when each repeat image reached the aircraft. All images of the aircraft after 18:22 UTC were taken in darkness with no Moon, so only the infrared channels could show a condensation trail until about 23:30 UTC.

Technical note: Aircraft at FL350 (10.67 km). Positions: Geoscience Australia digital military radar track 16:42:27-18:01:49 UTC (dashed); after that the core run J posterior (BTO+BFO), 148,000 routes, family weights waypoints 0.9982, free 0.0013, repro-radar 0.0003, descent-climb 0.0002, routes interpolated linearly between 600 s vertices; positions held at 00:11:49 for later scans. Satellite positions: SGP4 on the space-track element set nearest 20:00 UTC 7 Mar (Hugging Face mirror). Effective pixel size = nadir sampling x slant range / 35,786 km / sqrt(cos VZA) (area-equivalent); nadir sampling: IR 4 km (INSAT-3D, Elektro-L N1, COMS-1, MTSAT-2), 5 km (FY-2D/E, Meteosat-7), 8 km (INSAT-3A, Kalpana-1); visible 1, 1.25, 2, 2.5 km as listed in the sensor table (Kalpana-1 and INSAT-3A visible 2 km). Scan crossing: linear line-angle model per satellite; MTSAT-2 from HRIT line times (17:31:15.5-17:57:49.5 UTC, 0.580 s/line); others from published cadence, duration and direction (grades in the sensor table). FY-2D times are the scans held in the Chiba University archive (none 16:31-19:31). Sun and Moon: JPL DE421 at the weighted-median position; Moon 43 % illuminated, below the horizon throughout.